

MANAUS GOMES, Brazil

WMO: 821110

Lat: 3.039S

Lon: 60.050W

Elev: 264

StdP: 14.56

Time Zone: -4.00 (W04)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	71.2	71.5	64.4	91.4	90.9	66.3	97.8	88.4	15.8	79.8	13.2	82.9	1.8	350	0.291

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10	17.1	96.5	78.9	95.0	78.3	93.4	78.3	82.8	89.9	81.9	88.9	81.2	88.1	7.7	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
81.0	162.5	84.8	80.6	160.7	84.3	79.5	154.4	83.2	46.9	90.7	46.1	89.6	45.1	88.4	87.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
13.9	11.9	10.4	DB	69.4	98.3	1.6	1.9	68.3	99.7	67.4	100.8	66.5	101.9	65.4	103.3
			WB	67.8	83.3	1.9	2.2	66.4	84.9	65.3	86.2	64.2	87.4	62.8	89.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	81.6	80.7	80.6	80.8	80.7	80.8	81.2	81.7	82.8	83.2	83.3	82.6	81.3
	DBStd	2.67	2.71	2.53	2.38	2.33	2.16	2.37	2.24	2.23	2.64	2.79	2.57	2.65
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11552	953	858	954	920	956	936	982	1017	996	1033	977	970
	CDD65	6077	488	438	489	470	491	486	517	552	546	568	527	505
	CDH74	55040	3994	3505	3875	3714	4042	4325	4867	5631	5641	5913	5113	4420
	CDH80	21679	1332	1103	1207	1170	1307	1674	2078	2654	2674	2750	2141	1589
Wind	WSAvg	3.4	3.6	3.6	3.6	3.1	3.0	3.3	3.5	3.4	3.4	3.5	3.4	3.2
Precipitation	PrecAvg	90.1	11.2	11.2	12.8	12.6	9.6	4.6	3.0	2.2	2.9	4.5	7.2	9.4
	PrecMax	124.3	22.4	24.4	24.9	21.1	21.9	9.0	8.1	5.6	9.6	10.0	13.3	20.4
	PrecMin	69.4	1.3	4.3	5.3	5.4	2.6	0.4	0.1	0.1	0.0	0.4	0.5	3.5
	PrecStd	11.5	3.5	4.3	4.3	3.9	3.9	1.9	1.9	1.6	2.0	2.4	3.4	4.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	93.6	93.3	93.1	93.0	91.8	93.5	94.9	96.8	98.4	98.6	96.4	95.1
		MCWB	79.3	79.5	81.2	81.3	79.9	80.2	79.0	79.3	78.9	79.5	79.5	79.7
	2%	DB	91.4	91.3	91.0	90.0	90.0	91.5	93.0	94.9	96.1	95.4	93.6	92.8
		MCWB	79.1	78.8	79.5	79.1	79.0	78.7	78.3	78.0	77.9	78.2	78.7	79.0
	5%	DB	89.5	89.2	89.2	89.2	89.2	89.8	91.4	93.3	94.6	94.2	92.8	90.0
		MCWB	78.6	78.5	78.8	78.8	78.7	77.4	77.2	77.1	77.5	78.1	78.6	78.6
	10%	DB	87.4	86.3	86.3	86.3	87.4	88.1	89.6	91.5	91.8	91.8	90.0	88.0
		MCWB	78.3	78.1	78.2	78.2	78.6	77.4	76.9	76.9	77.5	78.3	78.4	78.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	82.4	82.8	83.0	83.3	82.8	82.8	82.3	82.5	82.7	83.0	82.8	82.7
		MCDB	88.0	88.7	89.3	89.3	88.4	89.8	89.5	91.4	92.7	92.6	90.3	89.4
	2%	WB	80.9	81.0	81.2	81.3	81.3	81.0	80.7	81.0	81.4	81.7	81.6	81.3
		MCDB	86.7	86.4	86.9	86.5	86.5	87.3	88.3	89.7	90.6	90.2	89.1	87.6
	5%	WB	80.0	79.9	80.0	80.1	80.2	80.0	79.8	80.1	80.5	80.7	80.7	80.5
		MCDB	85.3	84.8	85.2	84.9	85.2	85.8	86.5	88.2	89.1	88.9	88.2	86.4
	10%	WB	79.1	79.0	79.1	79.1	79.1	79.0	78.9	79.1	79.5	79.8	79.7	79.4
		MCDB	83.6	83.3	83.4	83.3	83.6	84.1	84.7	86.2	86.6	87.1	86.2	84.4
Mean Daily Temperature Range	5% DB	MDBR	13.1	12.8	12.8	12.9	13.1	14.7	16.1	17.8	17.9	17.1	15.7	13.9
		MCDBR	16.0	16.0	15.8	16.0	15.9	16.8	18.4	20.1	20.8	20.1	18.6	16.9
		MCWBR	6.1	6.1	6.2	6.4	6.2	6.1	6.1	6.0	5.9	5.9	6.0	6.4
	5% WB	MCDBR	14.2	13.9	14.4	14.2	14.2	15.5	16.8	18.3	18.6	18.2	16.5	15.0
		MCWBR	6.5	6.7	6.7	6.7	6.7	6.6	6.8	6.7	6.9	6.9	6.7	6.7
Clear-Sky Solar Irradiance	taub	0.399	0.394	0.393	0.385	0.387	0.391	0.403	0.486	0.526	0.505	0.492	0.451	
	taud	2.521	2.546	2.553	2.586	2.569	2.558	2.502	2.227	2.097	2.179	2.215	2.351	
	Ebn at Noon	297	298	295	290	281	276	274	258	254	263	267	279	
	Edh at Noon	35	35	34	32	31	31	33	45	53	49	47	41	
All-Sky Solar Radiation	RadAvg	1366	1352	1360	1334	1315	1457	1562	1683	1681	1605	1481	1331	
	RadStd	114	109	101	52	84	89	78	96	127	88	100	112	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		-0.40	-0.20	-2.93	-0.94	-2.07	-2.27	N/A	N/A	-275
(47) Regional (1 neighbor)		-0.40	-0.20	-2.93	-0.94	-2.07	-2.27	N/A	N/A	-275

Nomenclature: See separate page